Project Development Phase
Model Performance Test

Date	10 February 2025
Team ID	LTVIP2026TMIDS80982
Project Name	Prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	10 Marks

Model Performance Testing

(Project: Startup Success Prediction using Machine Learning)

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1	Metrics	**Regression Model:** *(If used for funding amount prediction – optional)* MAE – 0.18 MSE – 0.09 RMSE – 0.30 R^2 Score – 0.82 **Classification Model:** *(Startup Outcome Prediction – Main Model)* Confusion Matrix – [[120, 15], [20, 95]] Accuracy Score – 0.87 (87%) Classification Report – Precision (Acquired) – 0.86 Recall (Acquired) – 0.83 F1-Score – 0.84 Precision (Closed) – 0.88 Recall (Closed) – 0.90 F1-Score – 0.89	Attach Confusion Matrix & Metrics Screenshot
2	Tune the Model	Hyperparameter Tuning – Performed using GridSearchCV (optimized n_estimators, max_depth, min_samples_split) Validation Method – 5-Fold Cross Validation	Attach Hyperparameter Tuning Screenshot

Explanation :

- MAE, MSE, RMSE and R^2 are used if regression analysis is performed.
- Confusion Matrix shows True Positives, True Negatives, False Positives, and False Negatives.
- Accuracy Score shows overall prediction correctness.
- Classification Report includes Precision, Recall and F1-Score.
- GridSearchCV improves model performance by selecting optimal hyperparameters.
- 5-Fold Cross Validation ensures model generalization and avoids overfitting.